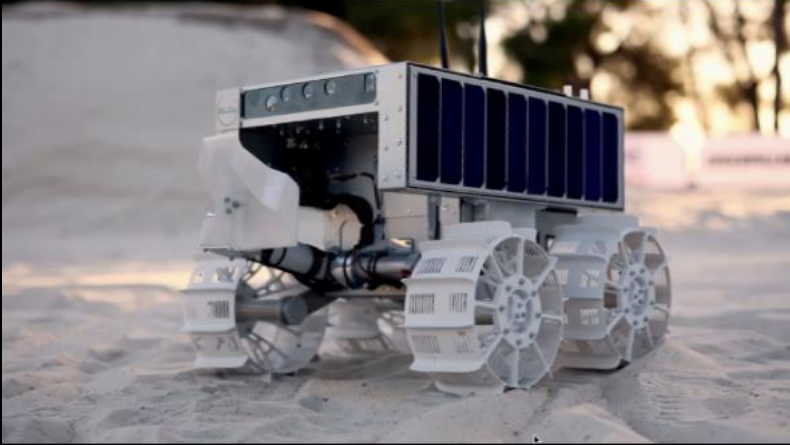


Towards Resilient Critical Infrastructures

Dr. Ahmad Mohsin

Advanced Manufacturing, Digital Twins and AI

Selected current projects



We contribute to the security of Australia's Lunar Roo-ver mission as part of the ELO2 Consortium.



We develop innovative AI-driven Digital Twins for resilient future smart manufacturing systems, focusing on cyber-physical systems security



As part of Australian Research in Power Systems Transition we develop security technology for Australia's EVs charging infrastructure. We also work with iMove on securing CITS that underpin modern traffic management.

Augmenting Cyber Defence Capability (ACDC)



Industry Engagement & Training



We work closely with domestic and international industry partners to translate research into real-world impact through collaborative projects, innovation, and sector-specific training. This includes initiatives such as the Cyber Resilience Fellowship: Strengthening Maldives–Australia Security Partnership, funded by the Department of Foreign Affairs and Trade (DFAT) Australia. We also integrate industry engagement into teaching through applied student projects, work-integrated learning, and industry-informed curriculum.

Towards Secure Industry 5.0 using AI, Digital Twins, and Intelligent Cyber Defense

Threats To Critical Infrastructure



EMERGING CYBER THREATS



Ransomware



OT/ICS Attack



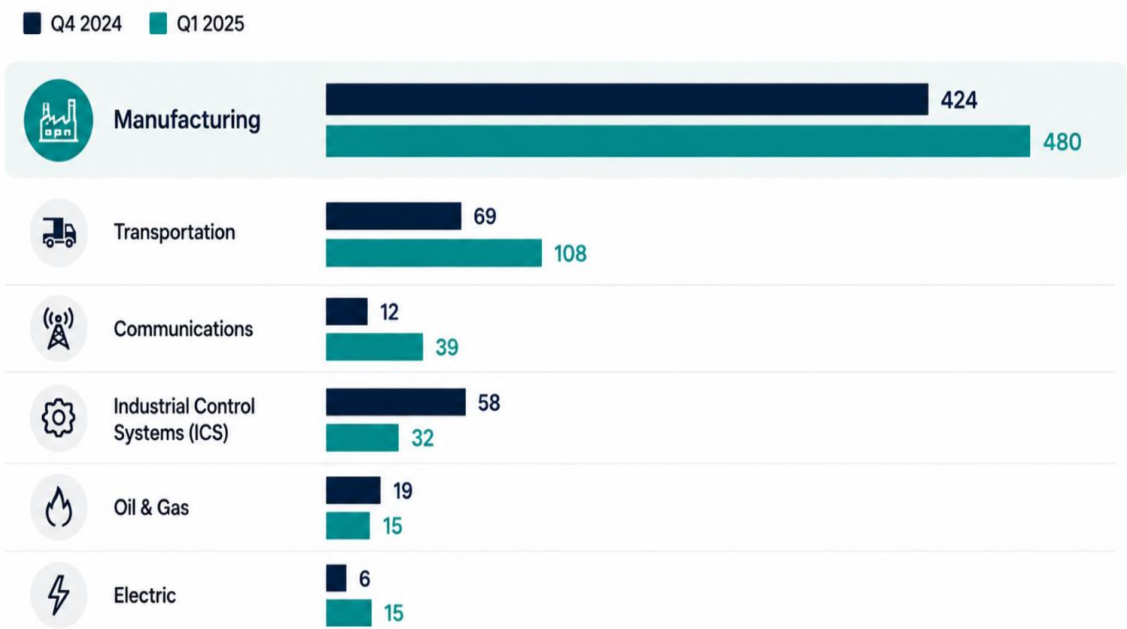
Supply Chain Intrusion



AI-Driven Threats

Ransomware Activity Across Critical Infrastructure

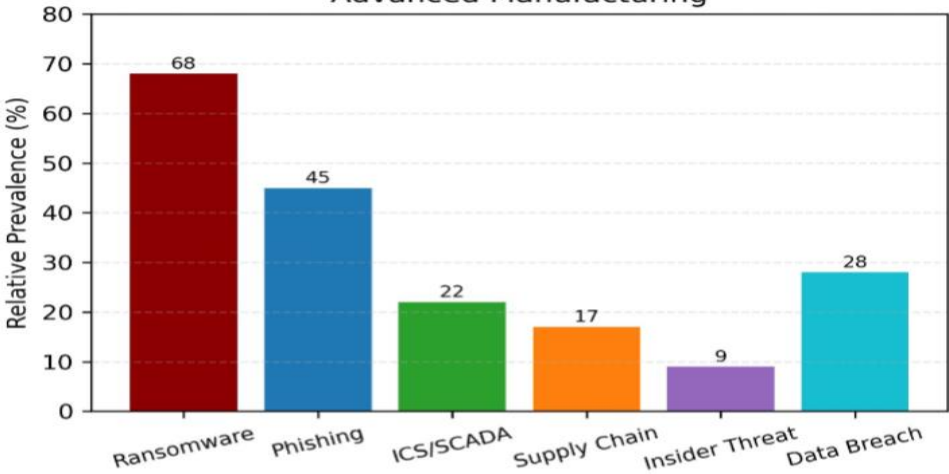
Ransomware by ICS Sector Q4 2024 vs Q1 2025



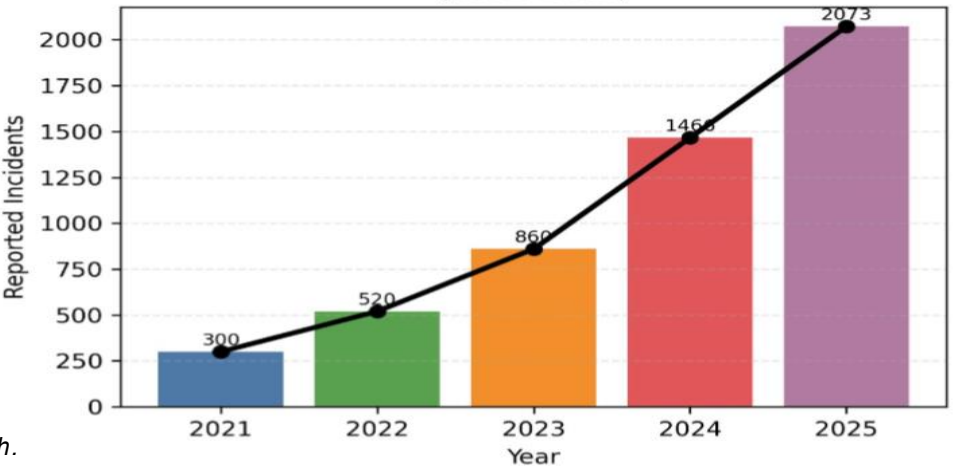
KEY INSIGHT

Industrial cyber attacks are rising rapidly as Manufacturing becomes more connected and automated

Common Cyber Attacks on Advanced Manufacturing



Increasing Industrial Cyber Attacks (2021-2025)



Smart Manufacturing

A Connected, Intelligent and Automated Manufacturing Environment



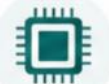
Robotic Systems



Conveyor & Transportation



Sensors & Actuators



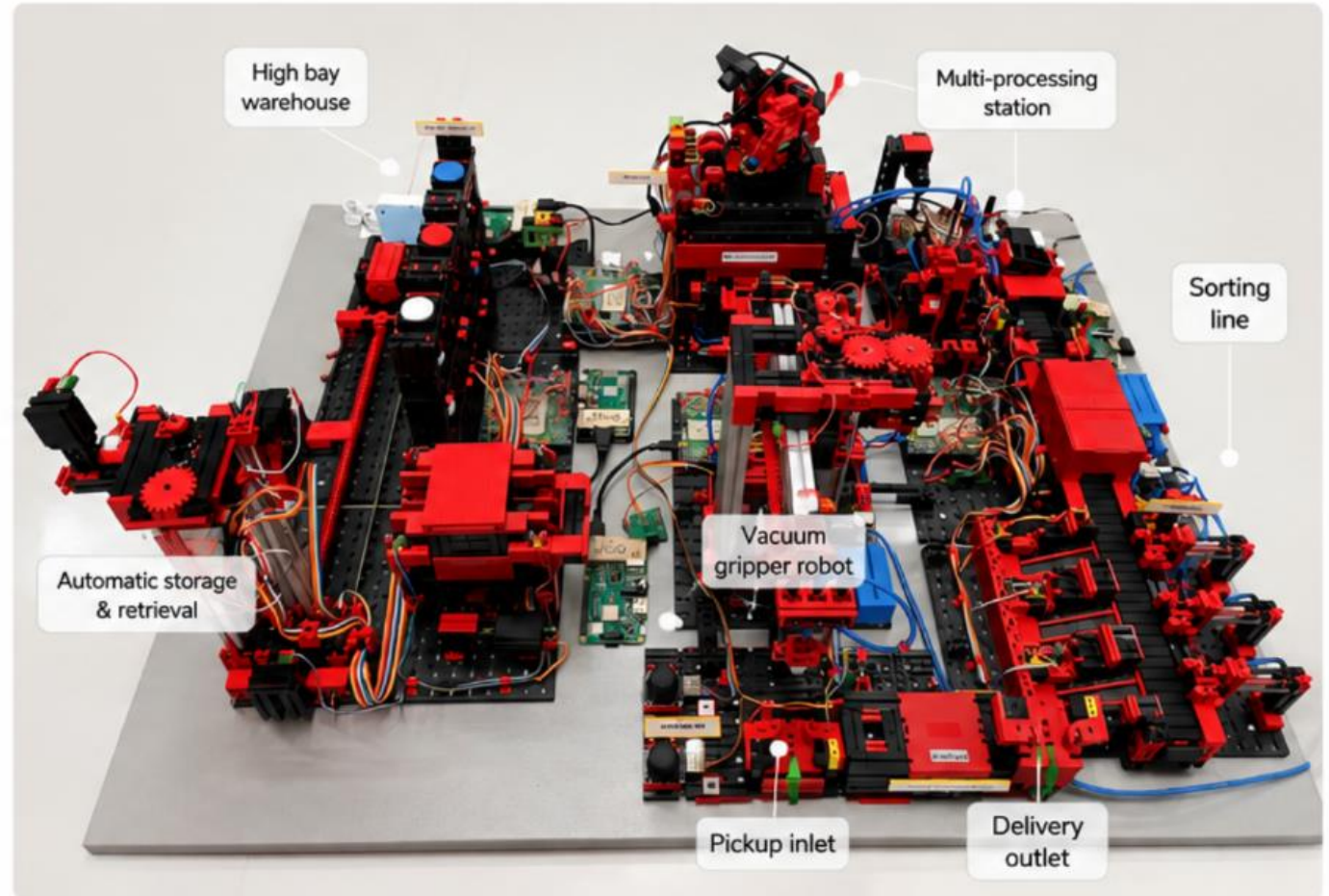
PLC Controllers



Automated Storage



Smart Sorting Systems



Defence Relevance and Strategic Impact

Aligning Industry 5.0 Cybersecurity Research with Australia Defence and Sovereign Capability Priorities



Critical Infrastructure Protection

Securing OT/ICS manufacturing systems, industrial automation, and cyber-physical infrastructures against advanced cyber threats.



AI-Driven Defence & Cognitive Security

Applying Agentic AI, Digital Twins, and Neuro-Symbolic Reasoning for intelligent threat detection and mission-aware cyber defence.



Sovereign Industrial Resilience

Supporting resilient Industry 5.0 manufacturing ecosystems aligned with sovereign capability, AUKUS, and advanced defence manufacturing.



Human-AI Collaboration for Defence Operations

Enhancing cyber situational awareness, attack-path analysis, and decision support for analysts, operators, and defence stakeholders.

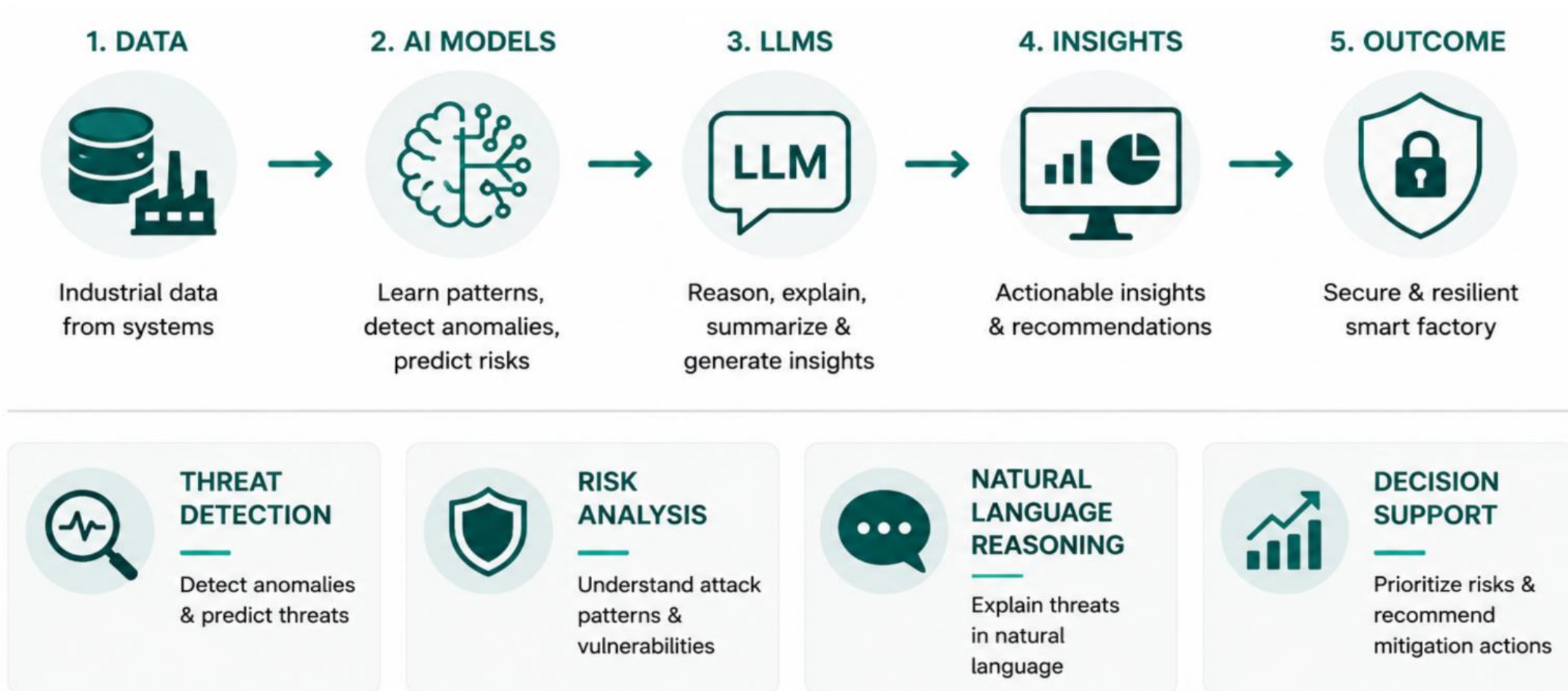


Research aligned with Defence priorities involving secure autonomous systems, critical infrastructure resilience, trusted AI, and Indo-Pacific cyber security.



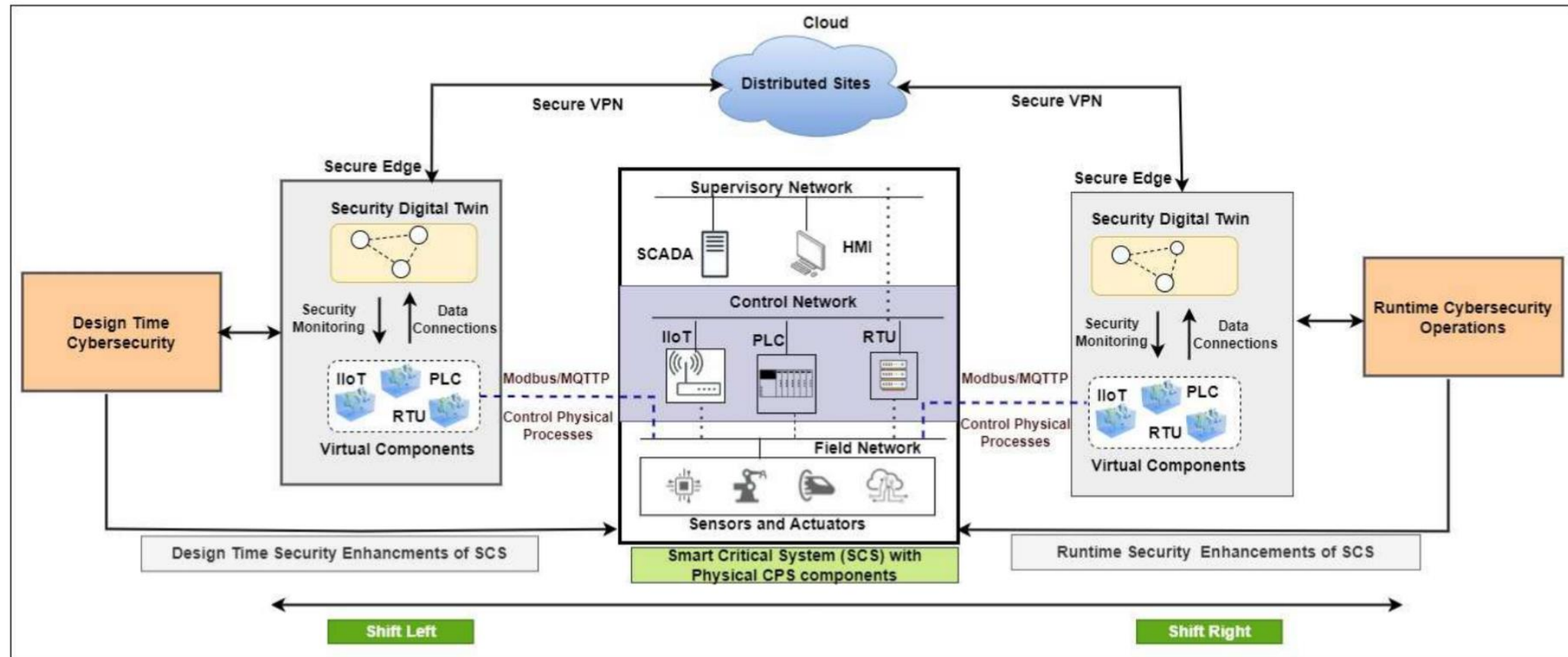
Why AI and LLMs?

Intelligent Technologies for smarter & more resilient manufacturing.

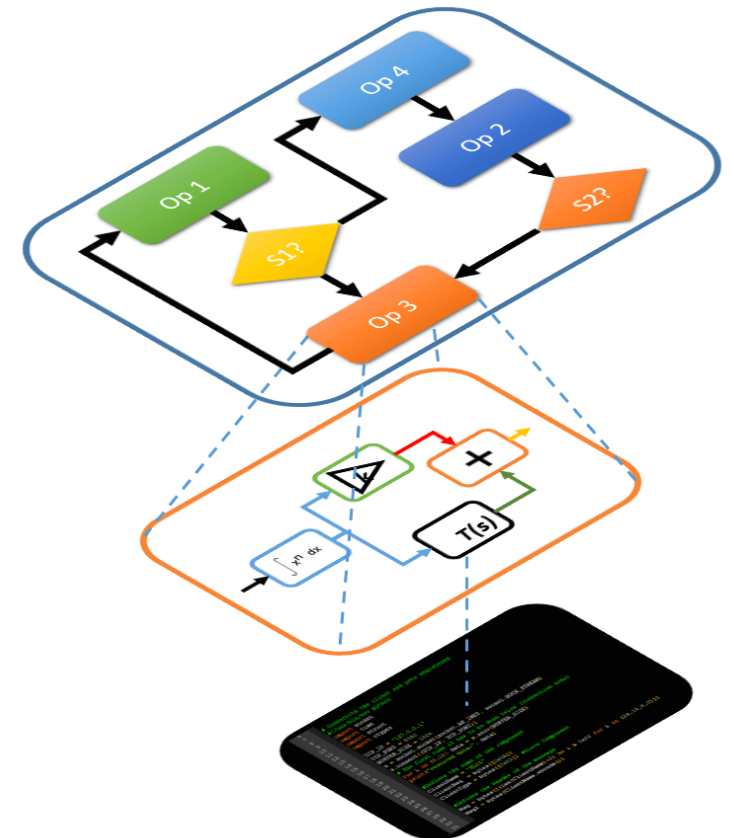


Digital Twins for Systems Resilience

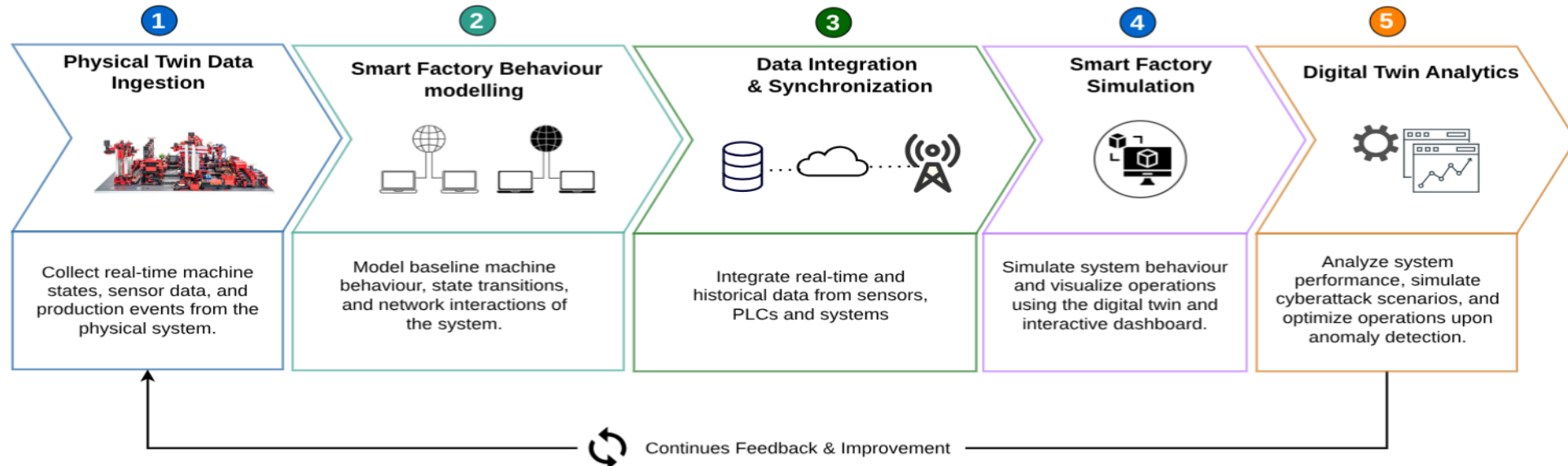
Digital Twins for CIs across the life cycle of Systems: Shift Left and Shift Right



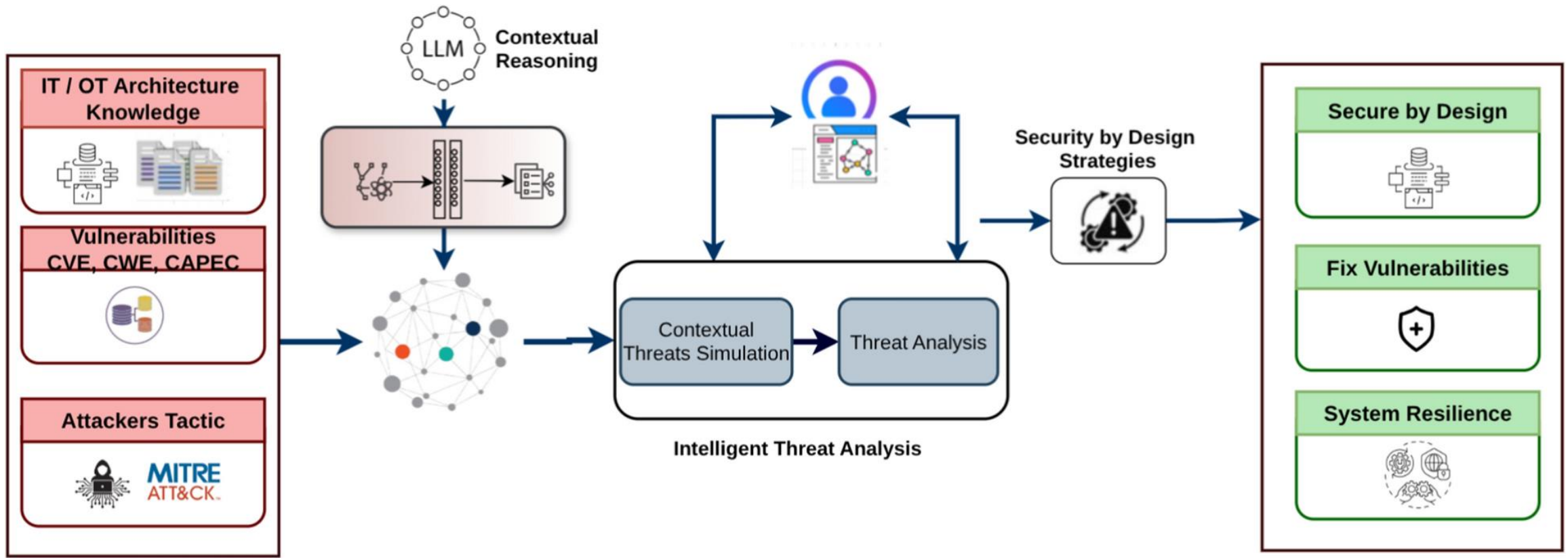
Digital Twins for Advanced Manufacturing



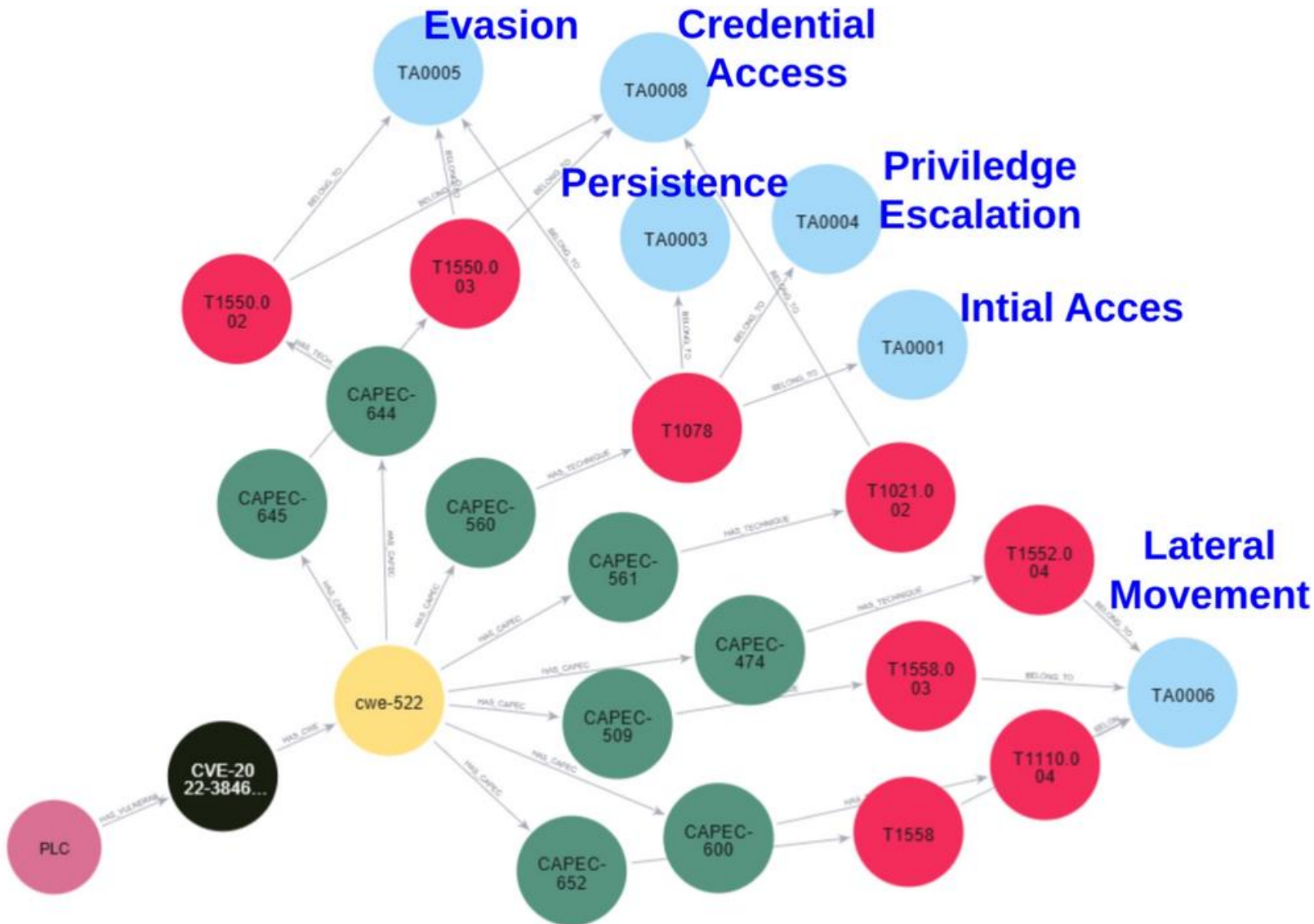
Digital Twins for Advanced Manufacturing



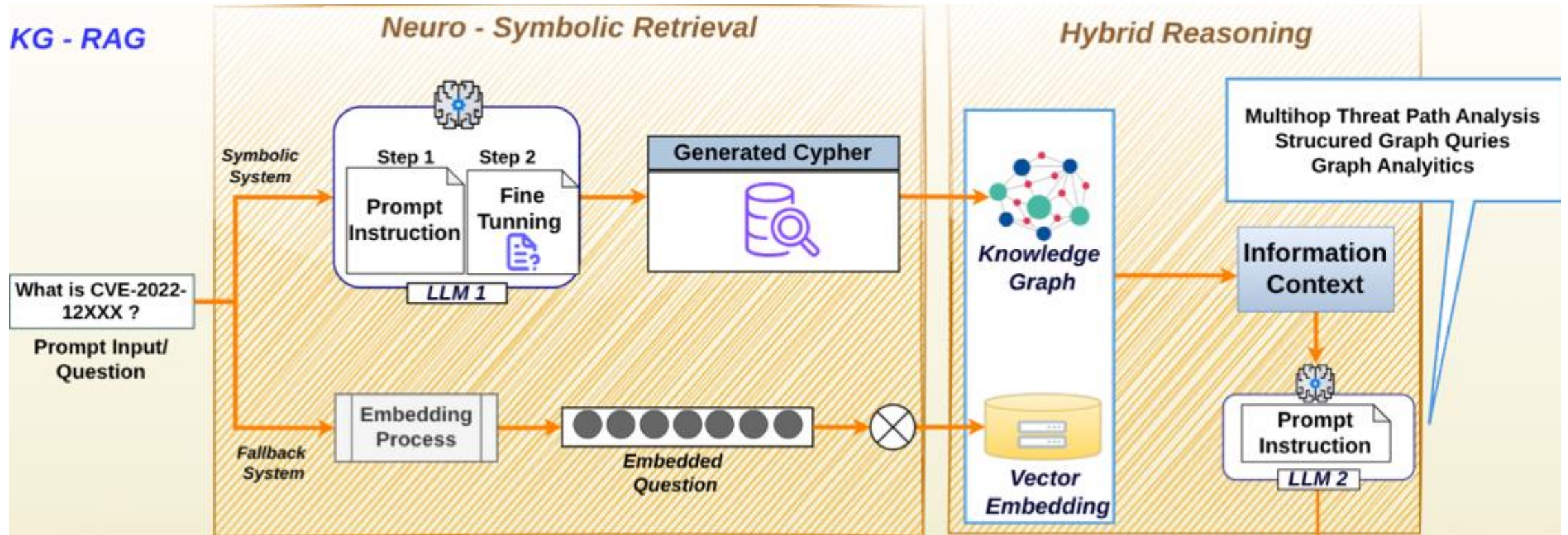
AI-Driven Threat Modelling- Knowledge Graphs



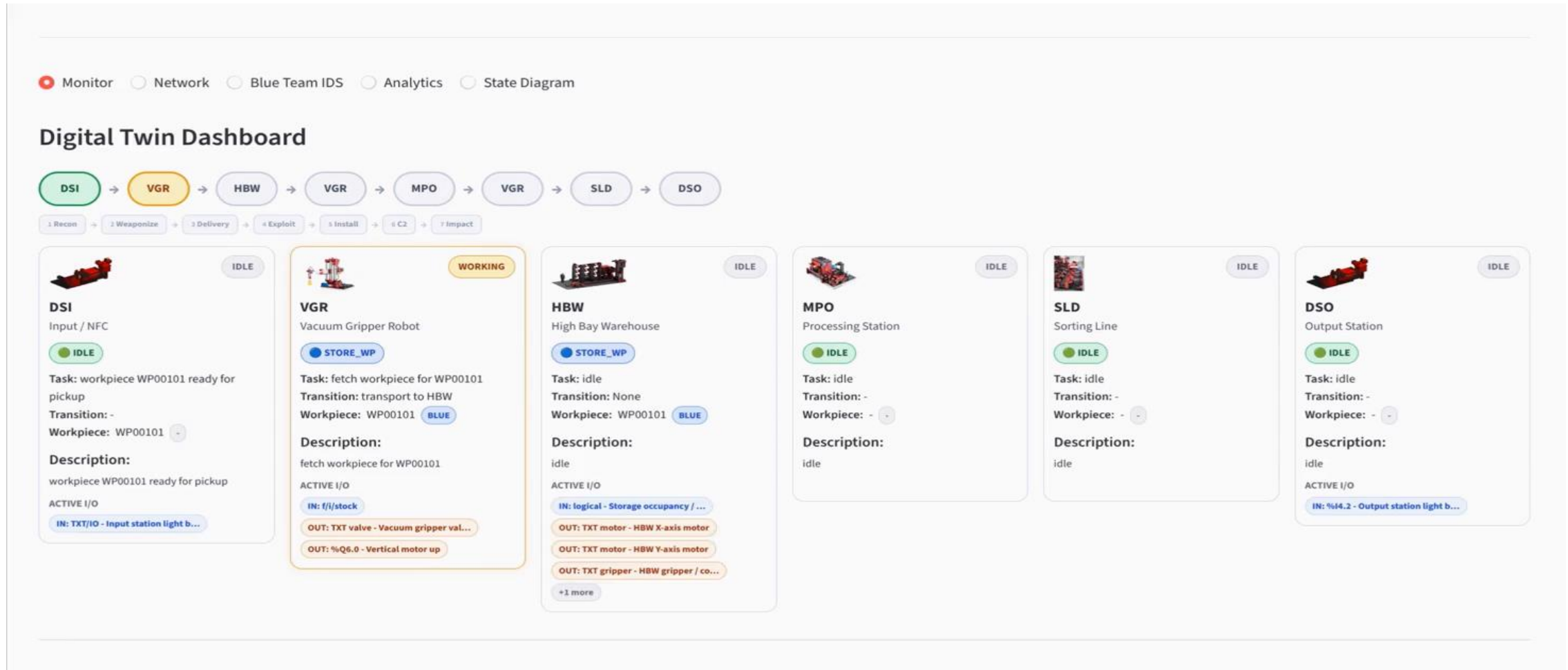
AI-Driven Threat Modelling- Knowledge Graphs



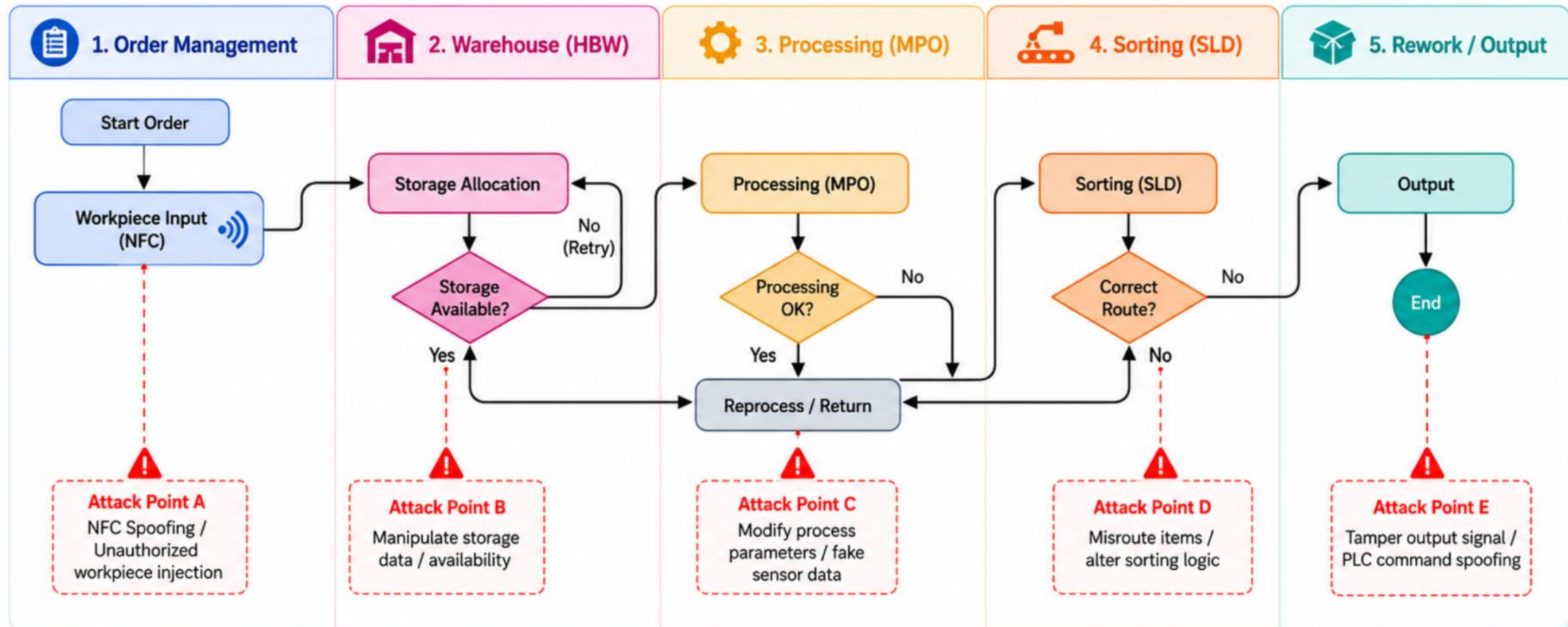
GRICS: A Human-Centric Neuro-Symbolic Knowledge-Graph Framework for Threat Reasoning in Advanced Manufacturing Systems.



Cyberattack Simulation via Digital Twin



Business Process & Example Attack



Work with us



Work with us to **support your organisation's strategic roadmap** on key security issues with state-of-the-art research.

Collaborative grants to support innovation.

Contract research to develop new IP for your organisation's future growth.

Support foundational research and **support Australia's research leadership** in security.

Develop your Talent Pipeline:

Engage with our students through projects and Work Integrated Learning.

Build skills through research and embed early career researcher and PhD candidates in your organisation or **retain existing staff** with the opportunity for further research studies.

Transfer of Knowledge and Research Translation:

Work with us to **understand pressing issues** relating to our research themes in depth.

Gain value and market share by commercialising joined IP.

Let us help you through targeted interventions and consultancy.

Get in Touch